Deployment Precision Calibration

Frozen HYB-MS +10% ensemble • identity-disjoint calibration only

GATE FAILED

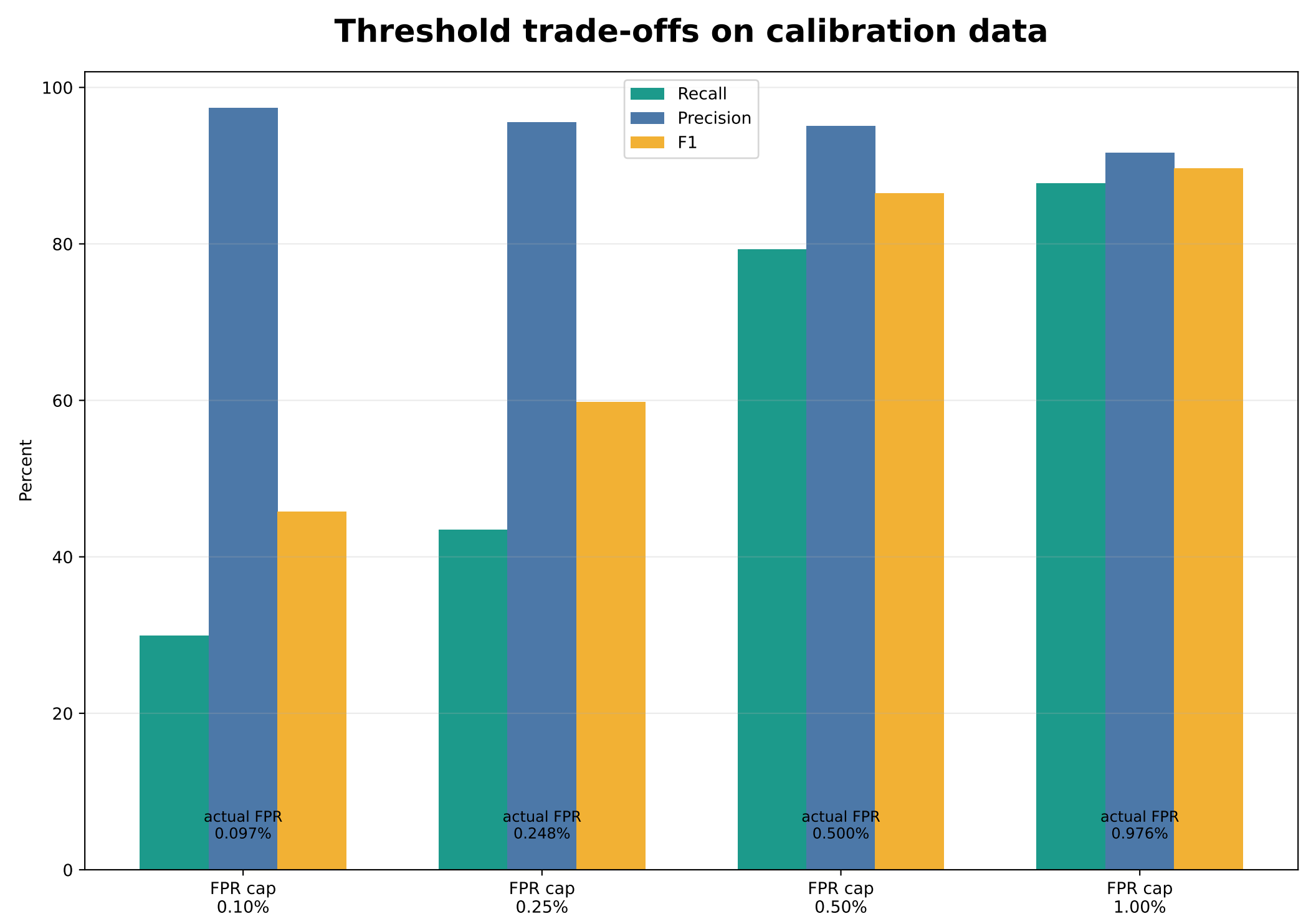
Selected threshold: 0.999849

Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
93.66%	95.52%	43.48%	59.76%	98.03%	92.71%	0.25%	56.52%

Gate: FPR ≤ 0.25% and recall ≥ 80%. Observed: FPR 0.25%, recall 43.48%.

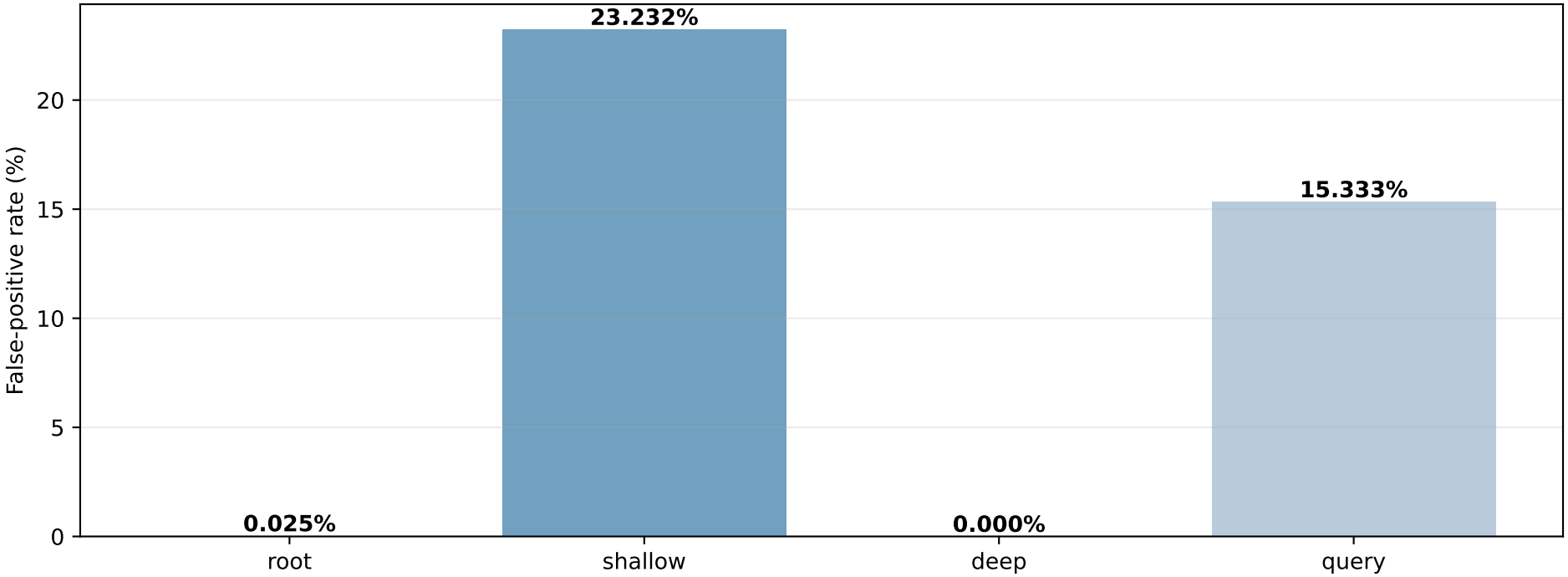
Counts: TN 20,549 • FP 51 • FN 1,413 • TP 1,087

This threshold was chosen without using the final confirmation set. If the gate fails, the model must be retrained using new development data and recalibrated on a separate cohort.



Original threshold 0.999512: recall 55.52%, FPR 0.34%. Selected deployment threshold is 0.999849.

Where False Positives Occur



Feature-pattern counts among false positives

query string: 23 • long url: 15 • none of registered flags: 26 • punycode: 1 • percent encoding: 1

These categories identify likely collection or representation weaknesses; they are diagnostic evidence, not permission to tune against the untouched confirmation set.

Operational Interpretation

At 0.1% phishing prevalence	**projected precision 14.95%**
At 1% phishing prevalence	**projected precision 63.95%**
At 5% phishing prevalence	**projected precision 90.24%**
At 10% phishing prevalence	**projected precision 95.13%**

Retraining decision

Gate failed; threshold adjustment alone is insufficient. Retrain with newly collected, structurally diverse legitimate development rows, then recalibrate on a separate unused cohort.

Next locked action

Apply this exact ensemble and threshold once to a newly frozen, identity-disjoint low-prevalence confirmation cohort. Do not revise the threshold after viewing those labels.